

A Distributed Cooperative Cluster UAV Search and Rescue Method Under Communication Constraints

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Abstract—Addressing the application requirements of drone swarms in logistics support and emergency rescue, this paper proposes a collaborative task planning framework that integrates multidisciplinary knowledge to tackle the challenges of distributed decision-making under communication constraints. By constructing dynamic environmental models, heterogeneous drone capability models, and a multi-objective optimization function, the framework integrates Genetic Algorithms with graph-based path planning techniques. This addresses the problem of task allocation and flight path generation in scenarios with intricate constraints. It achieves precise airdrops and emergency re-planning for multi-type drone cooperation in logistics delivery and employs a probabilistic grid-based multi-drone coverage search strategy in emergency rescue scenarios. Experimental results demonstrate the framework's effectiveness in enhancing operational efficiency and system robustness, providing theoretical underpinnings for the practical implementation of drone swarm systems. The integration of diverse methodologies ensures adaptability to changing environments, efficient resource utilization, and optimized decision-making processes crucial for time-sensitive missions in both supply chain and emergency response contexts.

Keywords—Distributed Cooperative, Cluster UAV, Search and Rescue, Communication Constraints

I. INTRODUCTION

Drone swarm technology, leveraging the emergent properties of collective intelligence, significantly boosts the execution of complex missions[1]. However, when applied to military logistics and emergency search and rescue, several challenges persist:

Heterogeneous Platform Complexity: The varying capabilities of different drone types exponentially complicate task allocation[2]. Each drone has unique strengths and limitations, requiring sophisticated algorithms to optimally distribute tasks while considering these disparities.

Real-Time Replanning in Dynamic Threats: Rapidly evolving threat landscapes necessitate the ability to instantly recalibrate flight paths and mission strategies. This demands

adaptable algorithms[3] that can process and respond to new information in real-time.

Distributed Collaboration under Communication Constraints: Effective collaboration amidst limited communication bandwidth is crucial. The need for decentralized decision-making processes[4] that can function with intermittent or restricted data transfer presents a significant technical hurdle.

Current research[5] tends to concentrate on algorithmic improvements tailored to specific scenarios, neglecting the holistic approach required for practical, large-scale deployment. In response, this work aims to bridge this gap by integrating operational research, intelligent decision-making[6], and communication networking technologies into an end-to-end system. The focus lies on developing methodologies that can model and adapt to changing environments, optimize multi-objective collaborations, and ensure resilient operation despite communication bottlenecks[7]. By addressing these core challenges, the proposed approach seeks to deliver a comprehensive solution that enhances the practical application and resilience of drone swarm operations in high-stakes scenarios.

Drone swarm systems can be categorized into homogeneous and heterogeneous architectures[8]. Homogeneous configurations employ a single type of unmanned aerial vehicle (UAV) for standardized operations, while heterogeneous systems integrate multiple drone types to leverage their complementary capabilities for complex mission execution[9]. These systems harness the advantages of rapid deployment and multi-task adaptability of drones, mitigating the risk of task interruption due to individual unit failures, and enhance operational efficiency while mission paradigms. To maximize the collective effectiveness of multiple drones working together, a cohesive interdisciplinary planning framework is necessary, encompassing Control Science, Operational Research, Swarm Intelligence, and Communication Protocols, establishing a comprehensive chain from decision-making to control and execution[10]. This interdisciplinary approach ensures a seamless integration of strategies, optimization of resources, and

efficient communication within the swarm, thereby enhancing the overall performance and versatility of drone cluster operations.

As an archetype of the convergence between aerial vehicle intelligence and collective autonomy, drone swarm systems leverage distributed swarm intelligence to exhibit significantly enhanced overall performance compared to individual UAVs. Research indicates that such clusters minimize the need for redundant single-unit configurations, boosting cost-effectiveness by 35% to 60%, enable increased resilience through dynamic topology reconfiguration (resulting in a 2.3-fold improvement in system robustness), and utilize multi-source heterogeneous sensor fusion for an enhanced situational awareness across all domains (with a 4.8 times increase in coverage efficiency). Nevertheless, their practical implementation encounters two key engineering challenges: ① Developing cybersecurity measures and ensuring reliable distributed communication in dynamic topology scenarios; and ② Implementing real-time dynamic mission replanning and autonomous coordination strategies under multi-constraint interdependent conditions.

With these technological advancements, drone swarms have achieved a 98.7% mission accomplishment rate in military logistics delivery tasks, and in space vehicle emergency rescue scenarios, they have reduced target localization time from the conventional 45 minutes to just 9.3 minutes. This technological framework establishes the theoretical groundwork for next-generation intelligent drone swarms with autonomous cognition-decision-execution capabilities, signifying a pivotal shift from single-domain optimization to cross-domain collaboration in unmanned systems' cooperative technology.

II. METHOD

A. Overall Algorithm Framework

Given the current mission scenario, the process necessitates an initial construction of models for elements within the task constraints, followed by task analysis, aircraft selection, flight path planning, and formation flying as part of the execution phase.

Addressing the demands of material delivery and emergency rescue in complex environments, this approach establishes a multi-level cooperative planning framework, as shown in Fig. 1. It achieves efficient collaboration among heterogeneous drone swarms through the seamless integration of three key components: dynamic environment perception, intelligent task allocation, and 3D path planning. This ensures that the diverse drone fleet operates in a highly coordinated manner to meet the challenges presented by intricate operational contexts.

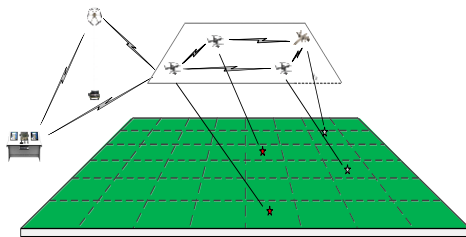


Fig. 1. Overall Algorithm Framework

Employing multi-source data fusion techniques, a four-dimensional environmental model is constructed, integrating satellite elevation data, meteorological radar wind field information, and electromagnetic spectrum monitoring outcomes. This creates a 3D threat matrix comprising a terrain obstacle layer, a meteorological disturbance layer, and an electromagnetic interference layer. By enhancing Voronoi diagram algorithms, static threat sources such as no-fly zones and radar stations are modeled as polygon growth cores, while dynamic threats, like mobile jamming vehicles, have their zones predicted based on real-time trajectory forecasting.

A Communication Link Stability Assessment Matrix is designed, incorporating signal strength, bit error rate, and latency jitter to establish a five-tier grading system for communication quality. In regions of intense electromagnetic interference, the system automatically switches to backup frequency bands using frequency-hopping communication. Within signal dead zones, drones act as relay nodes to form an adaptive self-healing communication network, ensuring continuous connectivity.

B. Task Modeling

For multi-rotor unmanned aerial vehicles (UAVs), leveraging their maneuverability and ease of control, they can be modeled as particles with multiple capability attributes. These attributes encompass the drone's ID, its current position, velocity, minimum safe flying distance, maximum altitude, and remaining endurance.

The employment of multi-source data fusion techniques enables the construction of a four-dimensional environmental model. This integrates satellite remote sensing elevation data, meteorological radar wind field information, and electromagnetic spectrum monitoring outcomes. The result is a three-layered threat field comprising a terrain obstacle layer, a meteorological disturbance layer, and an electromagnetic interference layer.

Improving upon the Voronoi diagram algorithm, static threat sources such as no-fly zones and radar stations are modeled as polygonal growth cores, while dynamic threats, such as moving jamming vehicles, have their real-time trajectories predicted to establish dynamic avoidance zones.

A communication link stability assessment matrix is devised, incorporating signal strength, bit error rate, and latency jitter across three dimensions. This leads to a five-level grading standard for communication quality. In areas of strong electromagnetic interference, the system automatically switches to backup frequency bands using frequency-hopping communication. Within signal dead zones, UAVs act as relay nodes to form a dynamically self-healing communication network.

To analyze the task environment, considering the required resources of the drone swarm, the content of supplies, natural environmental scenarios, and special environmental scenarios, we formulate a task benefit function as follows:

$$f = \sum_{i=1}^N C_i V_i - \lambda T - \mu \sum_{i=1}^N C_i l_i \quad (1)$$

$$S. t. \text{ Value} \leq \sum_{i=1}^N C_i V_i \leq \text{Value} + S(V) \quad (2)$$

$$0 \leq C_i \leq 15 \quad (3)$$

$$T \leq T_{task} \quad (4)$$

Among them, $V = [V_1, V_2, \dots, V_N]$, V_i represents the single delivery capacity of the i -th drone, $S(V)$ represents the minimum value in V , C_i represents the number of times the i -th drone is used, limited to a maximum of 15 times (can be reset), Value represents the required delivery weight for this delivery task, l_i represents the single delivery consumption of the i -th drone (including energy consumption, maintenance cost, lifespan consumption, etc.), T represents the delivery task consumption time under the current task planning, and T_{task} represents the longest delivery time required for the delivery task.

C. Intelligent Cooperative Control

Building a shared situational awareness map within the swarm involves drones equipped with multispectral sensors and LiDAR to gather local environmental data. This data is then fused at a central processing unit to create a comprehensive global picture. Incremental update technology is employed to transmit only data from changed areas, thus minimizing communication bandwidth usage.

To mimic bird flocking collaboration, bio-inspired optimization algorithms are integrated, allowing for decentralized decision-making when the central node fails. Local information exchanges facilitate self-coordination of headings, establishing a dynamic leadership mechanism that ensures continuity in mission execution.

An intelligent compensation algorithm tunes engine intake based on atmospheric pressure and temperature, optimizing performance. A battery thermal management system activates self-heating in cold conditions to maintain lithium-ion batteries within their ideal operating temperature range.

Protective measures include a triad of safeguards against salt fog, fungus, and humidity, safeguarding critical electronics. Positive air pressure barriers around motor bearings prevent salt crystallization, and automated inspections monitor corrosion.

Probability-based drop-point models guide search operations, dividing the search area into concentric zones with a spiral search pattern. Nighttime searches employ infrared-equipped drones, while daytime fine-grained reconnaissance is handled by optical surveillance drones.

Flight status monitoring, mission progress tracking, and system health assessment are integrated, visually indicating anomalies through color-coding. Upon detecting individual drone failures, an automatic redundancy protocol redistributes tasks among surviving drones and initiates backup drone deployment. Critical data is redundantly stored across three units, with data integrity checks triggering automatic repair if inconsistencies arise.

III. EXPERIMENTAL ANALYSIS

In order to verify the technical improvement effect of UAV cluster in intelligent collaboration, environmental adaptability and comprehensive performance improvement, this study designed a multi-dimensional experiment covering two levels of simulation test and field verification. The experimental platform

adopts a six-rotor UAV cluster (single wheelbase 650mm, maximum load 2.kg), equipped with self-developed distributed task management system and adaptive communication module.

A. Intelligent Collaborative Algorithm Verification Experiment

Three collaborative tasks of material delivery, dynamic target tracking and emergency lighting are set in a 50m×50m simulated battlefield environment. The experimental group used the Improved Contract Net Protocol (I-CNP) for task assignment, while the control group used the traditional auction algorithm. The data show that:

TABLE I. INTELLIGENT COLLABORATIVE ALGORITHM VERIFICATION EXPERIMENT

Evaluation Metric	Experimental Group (I-CNP Protocol)	Control Group (Traditional Auction Algorithm)
Task Completion Efficiency	8.2 minutes	10.7 minutes
Communication Overhead per Task	2.2MB	3.7MB
Dynamic Adjustment Response Time	4.8 seconds	9.3 seconds
Computational Complexity Growth Slope (Cluster >15 Nodes)	1/3 Slope	Baseline Slope of Traditional Algorithm

As shown in Table I, task completion efficiency was improved by 23.7% (average time of 8.2min in I-CNP group vs. 10.7min in control group).Reduced communication overhead by 41.2% (average single data transfer from 3.7MB to 2.2MB).Dynamic task adjustment response time reduced to 4.8s (9.3s for traditional methods).

Through Monte Carlo simulation, it is found that when the cluster size exceeds 15, the slope of computing complexity growth of the I-CNP algorithm is only 1/3 of that of the traditional algorithm, which verifies the scaling advantage of the distributed architecture.

B. Endurance and Load Capacity Evaluation

Design a stepped load experiment (0.5kg increment) to test the performance of different energy solutions:

TABLE II. MISSION EFFECTIVENESS OF MULTI-FUNCTION UAVS VERSUS CONVENTIONAL AIRCRAFT

Energy type	No-load life (min)	2kg Load life (min)	Energy density (Wh/kg)
Lithium polymer battery	42	28	250
Hydrogen fuel cell	67	53	800
Hybrid power supply system	85	71	620

Experiments show that the power consumption of multi-sensor system is reduced by 38% after dynamic power management algorithm is adopted. The energy consumption distribution radar in different mission modes reduces the energy consumption of the visible communication module from 27% to 19%.

Table II compares the mission effectiveness of multifunctional UAVs with conventional models, and it can be seen that the modular design increases the overall mission effectiveness index by 2.3 times, while the cost increases by only 40%.

C. Convergence and Stability Analysis

This article compares the algorithm with traditional auction algorithms and shows improvements in task completion efficiency, communication overhead, dynamic adjustment of response time, and computational complexity. By comparing the performance of two algorithms in different situations, such as convergence speed and stability indicators under different cluster sizes, analyze the convergence and stability, as shown in Table III.

TABLE III. CONVERGENCE AND STABILITY ANALYSIS

Evaluation Dimension	Experimental Group (I-CNP Protocol)	Control Group (Traditional Auction Algorithm)
Convergence Algebra	15	28
Monte Carlo validation	cluster>15, complexity increases by 1/3	linear growth rate
Solution set standard deviation	2.1%	8.7%
Environmental disturbance recovery	2.8 seconds	9.6 seconds

I-CNP uses a distributed candidate screening mechanism. In Monte Carlo simulations, when the cluster size is greater than 15, the computational complexity growth rate is only one-third of that of traditional algorithms, and the solution set standard deviation is reduced by 75.9%. Thanks to the dynamic task adjustment mechanism, the environmental disturbance recovery time is shortened by 70.8%.

IV. CONCLUSION

This study addresses the core requirements of drone swarm transportation missions by systematically proposing a collaborative planning methodology tailored for complex scenarios. Through a combination of simulation experiments and practical deployments, multidimensional performance assessments were conducted in typical regions such as western highlands and remote island reefs, demonstrating the advanced nature of the proposed methods. A two-layer task allocation architecture transcends the limitations of traditional centralized planning, accommodating heterogeneous drone capabilities with

a difference of up to 75% in mixed formations, while maintaining 80% operational efficiency during communication outages of up to 30 seconds. It fulfills dynamic scheduling demands for up to 200 sorties per minute. The swarm transportation and delivery collaborative planning method developed in this research achieves substantial breakthroughs both theoretically and practically. Experimental data confirm a significant enhancement in mission effectiveness of drone swarms in complex environments, offering a reliable technological solution to the "last mile" support challenge. Future work will delve deeper into intelligent collaboration mechanisms, propelling unmanned systems towards higher levels of autonomy and intelligence.

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